Ancheng Li

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Education

University of North Carolina at Chapel Hill — Chapel Hill, NC
Bachelor of Science in Computer Science and Statistics — *Aug. 2025 – Expected May 2029*

Nanjing University — Nanjing, China
Undergraduate Studies — *Aug. 2024 – May 2025*

Experience

Machine Learning Fellow — Aug. 2026 – Present
Automation Anywhere — *Remote*

- Fine-tuned Gemma 3 4B with LoRA to distill Claude Haiku 4.5 for domain-specific support-ticket labeling; filtered, cleaned and clustered 2,000 tickets with top 10 tickets selection and cluster-level splits to prevent data leakage.
- Developed a reference-free evaluation pipeline measuring faithfulness, specificity, and coherence; achieved a higher score than Claude Haiku 4.5—4.77/5 versus 4.67/5—across 50 test cases spanning 10 clusters.
- Increased throughput by 2.68x and reduced compute time by 62.7% (0.307 second/label) through batching; modeled AWS, Azure, and GCP costs, projecting a reduction from $0.33 to $0.0449 per 1,000 labels.

Business Intelligence and Data Analytics Intern — May 2026 – Aug. 2026
UNC Athletics Department — *Chapel Hill, NC*

- Processed and integrated NCAA athletics financial datasets spanning 2019–2024, combining with Directors' Cup standings datasets across 138 FBS institutions.
- Cleaned and standardized 1000+ institution-level records by resolving inconsistencies and creating custom fields in Tableau to support cross-dataset joins and analysis.
- Built Tableau visualizations of athletic spending and performance; identified UNC as the top-ranked institution in spending efficiency at 7.665 Directors' Cup points per $1M in expenditures and presented findings to stakeholders.

Break Through Tech AI Program — May 2026 – Aug. 2026
Cornell University — *New York, NY*

- Engineered an end-to-end NLP sentiment classifier across 2,000 text reviews, training custom Word2Vec embeddings and optimizing a Logistic Regression model to achieve a 0.72 ROC-AUC score.
- Developed a pharmaceutical RAG pipeline using ChromaDB and OpenAI's API, implementing 500-character chunking and top-5 similarity retrieval to reduce LLM hallucinations and increase answer accuracy.
- Designed multi-agent systems using LangGraph, OpenAI SDK. Implemented 8+ workflows incorporating key agentic design patterns to automate tasks like document analysis and report generation.

Projects

Automated Lecture Note and Transcript App | *Python, OpenAI API (Whisper, GPT-4o), FFmpeg* — July 2026

- Built an audio processing pipeline using FFmpeg to chunk 50+ MB lecture recordings into 10-minute segments, bypassing the 25 MB API upload limit.
- Automated speech-to-text conversion via OpenAI whisper-1, transcribing and aggregating 7 audio segments into a consolidated 174k+ character text corpus.
- Engineered system prompts for gpt-4o to synthesize unstructured transcripts into formatted study guides.

Handwritten Digit Classifier | *Python, TensorFlow, Keras, NumPy, Seaborn* — August 2026

- Built and trained a 4-layer Convolutional Neural Network with 99.4K parameters in TensorFlow/Keras to classify 10-class handwritten digits across 70,000 MNIST images.
- Designed a progressive feature extraction hierarchy ($16 \to 32 \to 64 \to 128$ filters, 3×3 kernels) integrated with Batch Normalization, ReLU activation, and Global Average Pooling to prevent overfitting.
- Preprocessed 28×28 grayscale arrays via 4D reshaping and $[0, 1]$ pixel scaling, achieving 94.09% test accuracy (0.203 test loss) using SGD optimization ($\text{lr} = 0.1$).

Technical Skills

Languages: Python, R, Java, C, SQL (Postgres), HTML, CSS
Developer Tools: Git, GitHub, Docker, Google Cloud Platform, Visual Studio Code, Rstudio, Tableau, IntelliJ, Jupyter Notebook, OpenAI API, Google Colab
Libraries: Pandas, NumPy, Matplotlib, Scipy, Tidyverse, Scikit-Learn, TensorFlow, Hugging Face Transformers